Config-Driven RL for Just-in-Time Adaptive Interventions

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Problem Definition

Goal: Optimise personalised physical-activity interventions with reinforcement learning.

MDP formulation: $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$

- $\mathcal{S}$: 36-state factored MDP (step bin, sleep quality, day of week, burden)
- $\mathcal{A}$: 4 actions (idle, movement suggestion, goal reminder, journal)
- $\mathcal{P}$: Unknown transition dynamics, estimated from LLM simulation or curated tables
- $\mathcal{R}$: $\alpha \cdot f(\text{step\_bin}) + (1-\alpha) \cdot g(\text{sleep}) - \lambda \cdot \mathbb{I}[\text{action} \neq \text{idle}]$

Core challenge: How do we iterate on RL agents without live patients?

Approach: Use a config-driven simulator with LLM-bootstrapped user models.

Existing Work

	HeartSteps V2	PEARL	StepCountJITAI
Type	MRT	4-arm RCT	Simulator
Sample	$n \approx 97$	$n = 13{,}463$	Synthetic
Algorithm	TS bandit	RL bandit	DQN
Actions	~5 types	155 nudges	4 types
Outcome	78.4% improved	+296 steps*	DQN $\approx$ 3000
Theory	SDT / FBM	COM-B	—

*RL vs control at 1 month, $p = 0.0002$.

Key gap: No existing framework combines *config-driven experiment definition* with *LLM-bootstrapped user simulation* for rapid agent prototyping before real-world deployment.

Our Approach

Two pillars:

1. **Config-Driven Framework**
   - Entire MDP defined in YAML, with no source-code changes
   - 36-state factored environment, expression-based rewards, 3 transition types
   - 7 RL/bandit agents, evaluation runner with multi-seed benchmarking
2. **Synthetic LLM Patients**
   - 22,320 prompts per persona across 5 behavioural personas
   - LLM-as-transition-model samples next-state probabilities
   - Total API cost: **$0.27** for the entire bootstrap

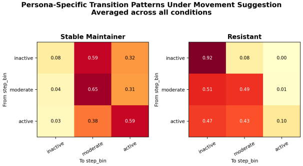
What We Have Done

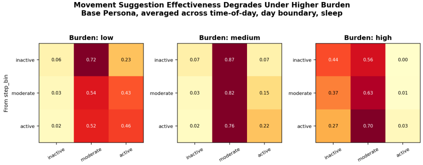
	MVP Simulator	LLM Bootstrap	Experiments
Purpose	Prove the pipeline	Generate patient behaviour	Evaluate agents
State space	2-state	36-state factored	Both
Transitions	Hand-curated rules	22,320 LLM prompts	From YAML config
Output	285 hyperparam combos	42 tables, 85 figures	7 personas $\times$ 7 agents
Cost	$0	**$0.27 total**	$0

Key point: A 2-state hand-curated MDP and a 36-state LLM-bootstrapped MDP run through the *same* YAML schema, same agent library, same evaluation runner.

Initial Results

Transition analysis — persona differentiation is real, burden degrades intervention effectiveness.





Persona	Optimal DP	HeartSteps (% opt)	Best Bandit (% opt)
Stable Maintainer	441.8	440.8 (99.8%)	402.5 (91.1%)
Base	379.7	379.1 (99.9%)	327.8 (86.3%)
Goal-driven	414.1	414.5 (100.1%)	323.0 (78.0%)
Social Responder	313.5	306.8 (97.9%)	140.1 (44.7%)
Initial DeepSeek	381.6	373.6 (97.9%)	297.5 (78.0%)
Initial GLM 5.2	374.6	374.0 (99.8%)	326.8 (87.2%)
Resistant	214.6	163.5 (76.2%)	151.7 (70.7%)

HeartSteps reproduced as a sanity check — 97–100% of optimal confirms the simulator is well-formed.

Next Steps

1. **Broader Agent Suite**
   - Add PEARL-style contextual bandit/RL agents for direct comparison
   - Stress-test offline RL, constrained RL, and adaptive bandit baselines
2. **Deep Learning Agents**
   - Benchmark DQN, actor–critic, and sequence-based policies
   - Explore representation learning for high-dimensional wearable state inputs
3. **More Complex Environments**
   - Expand state spaces with context, history, fatigue, and adherence dynamics
   - Model non-stationarity, delayed rewards, and intervention burden
4. **Sensor Foundation Models**
   - Wearable foundation-model embeddings for dynamic action spaces
   - Connect interventions to sensor-derived activity and sleep context
5. **LLM and Prompt Variants**
   - Compare DeepSeek, GLM, Qwen, Claude, and smaller local models
   - Test COM-B prompts, structured JSON, and multi-turn persona dialogues